

Advanced Spring Boot Assignment – Document Management System

Project Overview

Build a scalable Document Management System (DMS) backend using Spring Boot. The system should allow users to upload, manage, organize, version, and securely share documents. Focus on backend architecture, file handling, metadata management, and access control.

Business Scenario

Organizations need a centralized system to store and manage documents. Employees upload files, organize them into folders, share with others, and maintain versions.

Core Functional Requirements

- 1 User registration and authentication
- 2 Upload documents (PDF, images, etc.)
- 3 Download documents
- 4 Delete documents
- 5 Organize documents into folders
- 6 Store document metadata
- 7 Version control for documents
- 8 Search documents

Advanced Functional Requirements

- 1 Role-based access control
- 2 Share documents with other users
- 3 Track document access history
- 4 Soft delete and restore
- 5 Pagination and filtering
- 6 Document tagging system

Technical Requirements

- 1 Spring Boot 3
- 2 Spring Security with JWT
- 3 Spring Data JPA
- 4 MySQL/PostgreSQL
- 5 File storage (local/cloud)
- 6 Swagger documentation
- 7 Logging

Database Entities

- 1 User
- 2 Document
- 3 Folder
- 4 DocumentVersion
- 5 DocumentAccess

API Endpoints

- 1 POST /auth/register
- 2 POST /auth/login
- 3 POST /documents/upload
- 4 GET /documents/{id}
- 5 DELETE /documents/{id}
- 6 GET /documents
- 7 POST /folders
- 8 POST /documents/{id}/share

Evaluation Criteria

- 1 Code quality
- 2 Architecture
- 3 API design
- 4 Database design
- 5 Error handling

Deliverables

- 1 Spring Boot project
- 2 Database schema
- 3 Swagger docs
- 4 Postman collection
- 5 README